HIWI Programming Task

1. Background and Motivation

In Cooperative Perception, object detection pipelines often produce **false-positive detections**—objects that are detected but are not relevant to the current task. Being able to post-process and filter perception outputs is a common requirement for exchange of Collective Perception Messages (CPM).

For this exercise, a **ROS 2 bag file** is provided that contains real detection data from a camera-based perception system.

The bag file publishes messages on the topic: fused/detections with the message type (`vision_msgs/Detection2DArray`)

The detections include two object classes:

- **traffic_light** — a valid and relevant detection
- **bed** — an incorrect detection that must be filtered out

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2. Your Task

Your goal is to implement a ROS 2 node that performs real-time filtering of incoming detection messages.

The node must:

1. **Subscribe** to the topic: "fused/detections"
2. **Filter out** any detection whose class label contains the string: "bed"
3. **Publish** the cleaned detections on a new topic: "filtered/fused/detections"

The published message must:

- Use the same message type (`vision_msgs/Detection2DArray`)
- Preserve the original message header
- Preserve all fields of the remaining valid detections

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3. Expected Behavior

When the ROS 2 bag is played, the node should:

- Receive detection arrays from the original topic
- Remove all detections labeled as **bed**
- Republish only valid detections (e.g., **traffic_light**) to the filtered topic

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4. How to Verify Your Implementation

You can verify the correctness of your node by playing the provided bag file and monitoring the filtered topic:

ros2 topic echo filtered/fused/detections

No detection with the class label **bed** should appear in the output.

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5. Deliverables

Please submit the following:

- A **ROS 2 package** containing your node (implemented in either **Python** or **C++**)
- A short **README** file explaining:
  - how to build the package
  - how to run the node
  - how to test it with the provided bag file

Your node must be executable using:

ros2 run <package_name> <node_executable>

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6. Evaluation Criteria

Your submission will be evaluated based on:

- Correct use of ROS 2 publishers and subscribers
- Proper handling of message types and fields
- Code clarity and structure
- Completeness and clarity of documentation

Note the: Install the vision_msgs package for your ROS 2 distribution: sudo apt install ros-<distro>-vision-msgs to echo the message

Link for Rosbag: https://drive.google.com/file/d/1_e92RFXZXvwEY8tLp6nYyvgnRDb0A0Oa/view?usp=sharing

You are free to use any version of ROS2 distro....!!